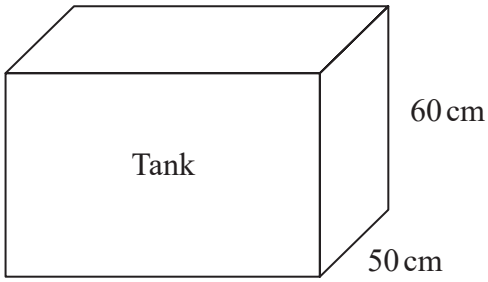
16





barrel

The diagram shows the dimensions of a tank in the shape of a cuboid.

The tank is $\frac{1}{3}$ full of water.

The diagram also shows a barrel that contains water.
Tina is told that there is 18 000 cm^3 of water in the barrel.
Tina is going to empty all the water from the barrel into the tank.

(a) What will the depth of water in the tank then be?

Base area = 100 × 50 = 5000 cm^2.

1 mark
Area of base

Initial water depth = 60 ÷ 3 = 20 cm.

1 mark
One-third full

Added depth = 18000 ÷ 5000 = 3.6 cm.

1 mark
Volume ÷ base area

Final depth = 20 + 3.6 = 23.6 cm.

1 mark
23.6cm

..... cm

(4)

Tina finds out that the barrel contains less than 18 000 cm^3 of water.

(b) Explain what effect this will have on your answer to part (a).

Less water added means a smaller added depth,

so the final depth will be less than23.6 cm.

1 mark
Final depth less

(1)

(Total for Question 16 is 5 marks)